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United States Patent Application Publication

20250279674

Kind Code

A1

Publication Date

September 04, 2025

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SIMULTANEOUS COMMISSIONING OF A PLURALITY OF SMART CIRCUIT BREAKERS IN A POWER DISTRIBUTION SYSTEM

Abstract

A method of simultaneously commissioning a plurality of smart circuit breakers to an energy control website in an IoT Hub in a cloud is provided. Each smart circuit breaker is structured to be coupled to a power source and respective load and provide circuit protection in a fault event. The method includes: forming, by the plurality of smart circuit breakers, a Wi-Fi network; commissioning a master smart circuit breaker to the energy control website in the IoT Hub via a mobile energy control application disposed in a mobile user device; simultaneously commissioning slave smart circuit breakers to the energy control website in the cloud via the mobile energy control application and the master smart circuit breaker; and connecting the master smart circuit breaker and the slave smart circuit breakers to the IoT Hub via an access point.

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Appl. No.: 18/594120

Filed: March 04, 2024

Publication Classification

Int. Cl.: H02J13/00 (20060101); G16Y30/00 (20200101); H04W12/06 (20210101); H04W84/12 (20090101)

U.S. Cl.:

Background/Summary

FIELD OF THE INVENTION

[0001] The disclosed concept relates generally to a system and method of securely executing commands via a cloud server, and in particular, to a system and method of simultaneous commissioning of a plurality of smart circuit breakers in a power distribution system.

BACKGROUND OF THE INVENTION

[0002] Cloud computing shares computing resources that are generally accessed via the Internet. Particularly, its infrastructure allows user to access a shared pool of computing resources, such as servers, storage devices, networks, applications, and/or other computing-based services. As such, the users and clients can remotely access computing resources on demand. IoT (Internet of Things) or edge IoT devices are physical electronic devices that are connected to a communication network, e.g., LAN, WAN, etc. and may be configured to transmit and receive information via the network. IoT devices can access cloud computing services or interact with an application via cloud servers and the network. IoT devices may be stationary, e.g., home appliance or manufacturing equipment, or mobile, e.g., cellular phones, vehicles accessories, etc. More and more devices are becoming IoT devices in consumer, industrial, energy, transportation, military and other fields.

[0003] However, as the IoT devices proliferate, issues rise with such proliferation. For example, many smart devices are configured with different set-up procedures (i.e., commission or provisioning procedures) depending on the type of the devices, capabilities of the devices or communication protocol employed by the devices. In general, one-to-one commissioning per device is required for the smart devices to be registered with the IoT Hub **16**. For example, when commissioning smart circuit breakers in a power distribution system, e.g., without limitation, one or more load centers, each smart circuit breaker undergoes one-to-one Wi-Fi commissioning with the IoT Hub **16** via a SoftAP (software enabled access point) and/or Bluetooth® Low Energy (BLE) Radio.

[0004] FIG. **3** illustrates the current one-to-one commissioning procedure for smart circuit breakers **1a-1n** installed in a power distribution system, where n is an integer. The one-to-one procedure includes five steps. At step 1, a first smart circuit breaker **1a** enters BLE mode and a mobile energy control application scans the first smart circuit breaker **1a**. In order to enter the BLE mode, a user actuates a switch or button **2a-2n** on a smart circuit breaker **1a-1n**. Upon actuating a switch **2a** on a first smart circuit breaker **1a**, the mobile energy control application start to scan its detection area (e.g., without limitation, 2-5 meters). At step 2, upon locating the first smart circuit breaker **1a**, the mobile energy control application **11** receives device ID of the first smart circuit breaker **1a**. The mobile energy control application **11** becomes paired to the first smart circuit breaker **1a** and authenticates the first smart circuit breaker **1a** via the BLE connection **4**. At step 3, the mobile energy control application transmits Wi-Fi credentials for an access point (AP) **14** to the first smart circuit breaker **1a** via the BLE connection **4**. The first smart circuit breaker **1a** in turn scans the Wi-Fi APs in the vicinity and transmits a list of the Wi-Fi APs to the mobile energy control application over the BLE connection **4**. The Wi-Fi APs appear on the mobile energy control application and the user selects the SSID of the AP **14** and enters a corresponding Wi-Fi credentials such as a password for the AP **14** into the mobile energy control application. Upon validation of the Wi-Fi credentials, the mobile energy control application is connected to the AP **14** and commissions the first smart circuit breaker **1a** to the energy control website in the cloud **5**. The mobile energy control application registers the first smart circuit breaker **1a** and configures the first smart circuit breaker

1a (e.g., without limitation, provide a name to the first smart circuit breaker 1a, etc.). At Step 4, the first smart circuit breaker 1a become indirectly connected to the IoT Hub 16 via the AP 14 and the user monitors the first smart circuit breaker 1a via the mobile energy control application. That is, the mobile energy control application uses the ID device address of the first smart circuit breaker 1a and connects the first smart circuit breaker 1a to the AP 14. The AP 14 in turn connects the first smart circuit breaker 1a via Ethernet to the IoT Hub 16. The mobile energy control application then disconnects the BLE connection 4 with the first smart circuit breaker 1a. At step 5, each of remaining smart circuit breakers 1b-1n undergoes the one-on-one commissioning procedure by repeating the Steps 1-4. The user can then monitor and control the smart circuit breakers 1a-1n within the power distribution system remotely via the mobile energy control application.

[0005] However, such one-on-one commissioning is not only time consuming, but also very costly, especially when the power distribution system includes a large number of smart circuit breakers to be commissioned.

[0006] There is room for improvement in commissioning IoT devices to IoT Hubs.

SUMMARY OF THE INVENTION

[0007] A method of simultaneously commissioning a plurality of smart circuit breakers to an energy control website in an IoT (Internet-of-Things) Hub in a cloud is provided. Each smart circuit breaker is structured to be coupled to a power source and respective load and provide circuit protection in a fault event. The method includes forming, by the plurality of smart circuit breakers, a Wi-Fi network, each smart circuit breaker being capable of acting as a master smart circuit breaker; commissioning a master smart circuit breaker to the energy control website in the IoT Hub via a mobile energy control application disposed in a mobile user device; simultaneously commissioning slave smart circuit breakers to the energy control website in the cloud via the mobile energy control application and the master smart circuit breaker; and connecting the master smart circuit breaker and the slave smart circuit breakers to the IoT Hub via an access point (AP).

[0008] Another example embodiment provides a method of automatically commissioning a replacement smart circuit breaker in a Wi-Fi network including a plurality of smart circuit breakers structured to be coupled to a power source and respective loads and provide circuit protection in a fault event. The plurality of smart circuit breakers have been commissioned simultaneously to an energy control website in an IoT Hub in a cloud upon forming the Wi-Fi network. The method includes: determining that the Wi-Fi network includes a faulty smart circuit breaker; in response to the determination that the Wi-Fi network includes a faulty smart circuit breaker, installing a replacement smart circuit breaker for the faulty smart circuit breaker; selecting a master smart circuit breaker from the commissioned smart circuit breakers and actuating the master smart circuit breaker; initiating, by the master smart circuit breaker, the automatic commissioning of the replacement smart circuit breaker; authenticating, by the replacement smart circuit breaker, the master smart circuit breaker and transmitting a device ID thereof to the master smart circuit breaker; transmitting, by the master smart circuit breaker, Wi-Fi credentials for an access point for the Wi-Fi network; authenticating, by the replacement smart circuit breaker, the Wi-Fi credentials; and connecting, by the replacement smart circuit breaker, to the AP and to the energy control website in the IoT using the Wi-Fi credentials.

[0009] Yet another example embodiment provides a system. The system includes a plurality of smart circuit breakers structured to be coupled to a power source and respective loads and provide circuit protection during a fault event; a mobile energy control application disposed in a mobile user device; an IoT Hub disposed in the cloud and communicatively coupled to the mobile energy control application, the IoT Hub including the energy control website; and an AP structured to be connected to the plurality of smart circuit breakers and the IoT Hub, wherein the plurality of smart circuit breakers are commissioned to the energy control website using a simultaneous Wi-Fi commissioning procedure. The simultaneous Wi-Fi commissioning procedure includes: forming, by the plurality of smart circuit breakers, a Wi-Fi network, each smart circuit breaker being capable

of acting as a master smart circuit breaker; commissioning a master smart circuit breaker to the energy control website in the IoT Hub via the mobile energy control application; simultaneously commissioning slave smart circuit breakers to the energy control website via the mobile energy control application and the master smart circuit breaker; and connecting the master smart circuit breaker and the slave smart circuit breakers to the AP and to the energy control website in the IoT Hub via the AP.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] A full understanding of the invention can be gained from the following description of the preferred embodiments when read in conjunction with the accompanying drawings in which:

[0011] FIG. 1 is a diagram of an exemplary system including a plurality of smart circuit breakers connected to an access point and a cloud in accordance with a non-limiting embodiment of the disclosed concept;

[0012] FIG. 2 is a sequential diagram that illustrates a method of simultaneously commissioning a plurality of smart circuit breakers in a Wi-Fi network in accordance with a non-limiting embodiment of the disclosed concept;

[0013] FIG. 3 is a sequential diagram that illustrates a conventional method of one-to-one commissioning of a plurality of smart circuit breakers;

[0014] FIG. 4A illustrates Step 1 of the method of simultaneously commissioning of FIG. 2. At Step 1 sets forth a process of forming a Wi-Fi network of a plurality of smart circuit breakers to be simultaneously commissioned to an IoT Hub in accordance with a non-limiting embodiment of the disclosed concept;

[0015] FIG. 4B illustrates Step 2 of the method of simultaneously commissioning of FIG. 2. At Step 2, a master smart circuit breaker is commissioned to an energy control website in the cloud in accordance with a non-limiting embodiment of the disclosed concept;

[0016] FIG. 4C illustrates Step 3 of the method of simultaneously commissioning of FIG. 2. At Step 3, the slave smart circuit breakers are simultaneously commissioned to the energy control website in the cloud in accordance with a non-limiting embodiment of the disclosed concept;

[0017] FIG. 4D illustrates Step 4 of the method of simultaneously commissioning of FIG. 2. At Step 4, all of the smart circuit breakers in the Wi-Fi network is connected to the IoT Hub via the AP in accordance with a non-limiting embodiment of the disclosed concept;

[0018] FIG. 5 is a flow chart for a method of simultaneously commissioning a plurality of smart circuit breakers in a Wi-Fi network in accordance with a non-limiting embodiment of the disclosed concept; and

[0019] FIG. 6 is a flow-chart for a method of replacing a faulty smart circuit breaker in the Wi-Fi network of FIG. 2 and commissioning a replacement smart circuit breaker in the Wi-Fi network in accordance with a non-limiting embodiment of the disclosed concept.

DETAILED DESCRIPTION OF THE INVENTION

[0020] Directional phrases used herein, such as, for example, left, right, front, back, top, bottom and derivatives thereof, relate to the orientation of the elements shown in the drawings and are not limiting upon the claims unless expressly recited therein.

[0021] As employed herein, the statement that two or more parts are “coupled” together shall mean that the parts are joined together either directly or joined through one or more intermediate parts.

[0022] FIGS. 1, 2, 4A-D and 5 illustrate a power system **100** including a plurality of smart circuit breakers **1a-1n** and a method **400,500** of simultaneously (substantially simultaneously) commissioning the plurality of smart circuit breakers **1a-1n** to an energy control website stored in the cloud **5** via a mobile energy control application **11** in accordance with a non-limiting

embodiment of the disclosed concept. The plurality of smart circuit breakers **1a-1n** may be installed in a power distribution system, e.g., without limitation, a load center **10**. The power system **100** may include a plurality of load centers **10**, each load center including a plurality of smart circuit breakers and associated with a single access point (AP) **14** or its own AP within respective Wi-Fi range (e.g., without limitation, 4-5 meters). The smart circuit breakers **1a-1n** are standard circuit breakers **1a-1n** structured to be coupled to a power source and a load and provide circuit protection during a fault event. The smart circuit breakers **1a-1n** can be remotely controlled via a mobile energy control application **11** installed in a mobile user device **12**. The smart circuit breakers **1a-1n** form a Wi-Fi network **103** using a wireless communication protocol such as ESP-NOW protocol. The ESP-NOW protocol is defined by Espressif and enables a direct and low-power control of IoT devices without using a router. The smart circuit breakers **1a-1n** then are simultaneously commissioned to the energy control website via the Wi-Fi connection **3**.

[0023] As shown in FIGS. **2**, **4A-D** and **5**, the simultaneous commissioning procedure of the plurality of smart circuit breakers **1a-1n** consists of four steps. As described with reference to FIG. **5**, the four steps include: forming, by the plurality of smart circuit breakers **1a-1n**, the Wi-Fi network **103**, each smart circuit breaker **1a-1n** being capable of acting as a master smart circuit breaker; commissioning a master smart circuit breaker **1a** to the energy control website in the IoT Hub **16** via the mobile energy control application **11** disposed in the mobile user device **12**; simultaneously commissioning slave smart circuit breakers **1b-1n** to the energy control website in the cloud **5** via the mobile energy control application **11** and the master smart circuit breaker **1a**; and connecting the master smart circuit breaker **1a** and the slave smart circuit breakers **1b-1n** to the IoT Hub **16** via the AP **14**.

[0024] FIGS. **4A-D** provide detailed sub-steps for each of the four steps. The detailed sub-steps of each broader step are provided below with reference to FIGS. **2** and **4A-D**. While the master smart circuit breaker **1a** and the slave smart circuit breakers **1b-1n** are commissioned substantially simultaneously, the simultaneous commissioning procedure includes simultaneous commissioning of slave smart circuit breakers **1b-1n** without having to undergo one-to-one commissioning currently required by the conventional commissioning procedure. That is, all of the slave smart circuit breakers **1b-1n** is commissioned at the same time, not one by one as currently required by the conventional commissioning procedure.

[0025] At Step **1**, the plurality of smart circuit breakers **1a-1n** form a Wi-Fi network **103**. Forming the Wi-Fi network **103** includes installing the plurality of smart circuit breakers in a power distribution system; actuating a switch on one smart circuit breaker to commence a Wi-Fi network forming procedure; transmitting, by the one smart circuit breaker, a request to connect; actuating switches on remaining smart circuit breakers; transmitting, by the remaining smart circuit breakers, responses to the one smart circuit breaker; and forming, by the one smart circuit breaker and the remaining smart circuit breakers, the Wi-Fi network based on the responses. All of the smart circuit breakers **1a-1n** has been installed in the load center **10** and is in a secondary mode (a slave mode). Each smart circuit breaker **1a-1n** includes a switch **2a-2n** and upon actuation of the switches **2a-2n**, the smart circuit breakers **1a-1n** form a wireless communication connection, e.g., without limitation, a Wi-Fi connection, a Bluetooth® Low Energy (BLE) connection, etc. The switch **2a-2n** may use the ESP-NOW protocol, which coexists with the Wi-Fi and BLE protocols. To form the Wi-Fi Network **103**, a user actuates the switch **2a-2n** on one smart circuit breaker **1a-1n** at **411**. Upon actuation of the switch **2a-2n**, the first smart circuit breaker **1a-1n** is placed in a primary mode and acts as a master smart circuit breaker in ESP-NOW pairing (e.g., without limitation, one-to-many). At **412**, the user promptly actuates the switches **2a-2n** on the remaining smart circuit breakers **2a-2n**. At **413**, the one smart circuit breaker **1a-1n** starts an ESP NOW responder. At **414**, the remaining smart circuit breakers **1a-1n** start an ESP-Now initiator as slave devices. Upon the ESP-NOW pairing, the smart circuit breakers **1a-1n** form the Wi-Fi network **103** using the ESP-NOW protocol. Upon forming the Wi-Fi network **103**, the remaining smart circuit breakers **1a-1n**

transmit MAC (media-access-controller) address including device IDs (UUID (universal unique individual identifier)) to the one smart circuit breaker **1a-1n** over the Wi-Fi connection **3**. At **415**, the one smart circuit breaker **1a-1n** then creates a device ID list of all of the smart circuit breakers **1a-1n** within the Wi-Fi network **103**.

[0026] At Step **2**, a user selects any one of the smart circuit breakers **1a-1n** to act as the master smart circuit breaker **1a** and the master smart circuit breaker **1a** is commissioned to the energy control website in the IoT Hub **16**. The commissioning the master smart circuit breaker **1a** includes: selecting a smart circuit breaker **1a** to act as the master smart circuit breaker; actuating a switch on the master smart circuit breaker **1a** and placing the master smart circuit breaker **1a** in a Bluetooth Low Energy (BLE) mode; scanning, by the mobile energy control application **11**, for presence of smart circuit breakers **1a-1n** in detection range thereof in the BLE mode; connecting, by the mobile energy control application **11**, to the master smart circuit breaker **1a** via the BLE connection **4**; authenticating Wi-Fi credentials of the AP **14**; connecting, by the mobile energy control application **11**, the energy control website in the IoT Hub **16** to the AP **14** based on the authentication; determining, by the mobile energy control application **11**, that the energy control website in the IoT Hub **14** is successfully connected to the AP **14**; and commissioning the master smart circuit breaker **1a** based on the determination that of the energy control website in the IoT Hub **16** is successfully connected to the AP **14**. Step **2** includes sub-steps **421-426**.

[0027] At **421**, the user actuates the switch **2a** of the master smart circuit breaker **1a** and places the master smart circuit breaker **1a** in a BLE mode. At **422**, the mobile energy control application **11** scans using a BLE radio and becomes paired with the master smart circuit breaker **1a**. At **423**, the master smart circuit breaker **1a** scans Wi-Fi APs in vicinity thereof, creates a list of the Wi-Fi APs, and transmits the list of the Wi-Fi APs to the mobile energy control application **11**. The master smart circuit device **1a** may also transmit a device ID list of all of the smart circuit breakers **1a-1n** within the Wi-Fi network **103** to the mobile energy control application **11**. The Wi-Fi APs in the vicinity appear on the mobile energy control application **11**. At **424**, the user authenticates the Wi-Fi credentials of the AP **14**. That is, the user selects an SSID (service set identifier) of the AP **14** and enters, e.g., without limitation, a password to the AP **14**. Upon authentication of the Wi-Fi credentials, the mobile energy control application **11** checks if the energy control website in the IoT Hub **16** is successfully connected to the AP **14**. If the Wi-Fi credentials fail, the master smart circuit breaker **1a** may reset and send a reset command to the slave smart circuit breakers **1b-1n** via the ESP-NOW. At **425**, upon successful connection of the AP **14** to the energy control website in the IoT Hub **16**, the master smart circuit breaker **1a** becomes successfully commissioned to the energy control website and appears on the mobile energy control application **11**. At **426**, the user can configure the master smart circuit breaker **1a** for use via the mobile energy control application **11**. For example, the user may provide a designation (e.g., without limitation, a kitchen circuit breaker) for the master smart circuit breaker **1a**. Further, the mobile energy control application **11** then creates a list of the smart circuit breakers **1a-1n** within its detection range (e.g., without limitation, 2-3 meters) and registers the smart circuit breakers **1a-1n** to the energy control website. The BLE connection may be turned off upon the configuration of the master smart circuit breaker **1a**.

[0028] At Step **3**, the slave smart circuit breakers **1b-1n** are simultaneously commissioned to the energy control website. Simultaneous commissioning of the slave smart circuit breakers **1b-1n** includes: receiving, by the slave smart circuit breakers **1b-1n**, the Wi-Fi credentials of the AP **14** from the master smart circuit breaker **1a**; and establishing, by the slave smart circuit breakers **1b-1n**, a connection to the IoT Hub **16** using a connection string and the Wi-Fi credentials via the mobile energy control application **11**. Step **3** includes sub-steps **431-433**. At **431**, the slave smart circuit breakers **1b-1n** receive the Wi-Fi credentials of the AP **14** from the master smart circuit breaker **1a**. At **432**, the slave smart circuit breakers **1b-1n** establish a connection to the IoT Hub **16** using a connection string and the Wi-Fi credentials via the mobile energy control application **11**. Upon successful connection to the IoT Hub **16**, the slave smart circuit breakers **1b-1n** become

simultaneously commissioned to the energy control website in the cloud **5**. At **433**, the slave smart circuit devices **1b-1n** appear on the mobile energy control application **11** and the user can configure the slave smart circuit breakers **1b-1n**. At **434**, if the mobile energy control application **11** fails to register device IDs of all of the smart circuit breakers **1a-1n** to the energy control website in the cloud **5**, the mobile energy control application **11** may reset the device ID list via the cloud **5** and the commissioning procedure is reinitiated by the user.

[0029] At Step **4**, the mobile energy control application **11** connects all of the smart circuit breakers **1a-1n** to the AP **14**, and thus indirectly connect the smart circuit breakers **1a-1n** to the energy control website in the IoT Hub **16** via the AP **14**. The user can monitor and control the smart circuit breaker **1a-1n** via the mobile energy control application **11** remotely.

[0030] In some examples, a replacement smart circuit breaker for, e.g., without limitation, a faulty smart circuit breaker may be automatically commissioned to the energy control website in the cloud **5** without having to undergo the BLE commissioning with the mobile energy control application **11**. FIG. **6** describes in detail adding a replacement smart circuit breaker in the existing Wi-Fi network **103** and commissioning the replacement smart circuit breaker.

[0031] In some examples, one or more additional smart circuit breakers may be added to the Wi-Fi network **103**. In those examples, Steps **1-4** are performed for commissioning the one or more additional smart circuit breakers.

[0032] Accordingly, the inventive simultaneous commissioning method **400,500** allows a plurality of smart circuit breakers **1a-1n** to first form a Wi-Fi network **103** without having to connect to the AP **14**. The inventive simultaneous commissioning method **400,500** also allows the plurality of smart circuit breakers **1a-1n** be commissioned simultaneously, thereby dispensing with the time-consuming and costly one-to-one commissioning procedure required under the conventional Wi-Fi commissioning methods. Further, a replacement smart circuit breaker or a new smart circuit breaker may be automatically commissioned without requiring a field personnel's visit, thereby saving installation costs and time. In addition, if one or more smart circuit breakers are relocated, the user can simply update Wi-Fi credentials of a new AP for the relocated smart circuit breakers, further increasing the efficiency and convenience to the user.

[0033] FIG. **5** is a flow chart for a method **500** of simultaneous Wi-Fi commissioning of a plurality of smart circuit breakers in a power distribution system in accordance with a non-limiting, example embodiment of the disclosed concept. The method **500** may be performed by the components of the system **100** of FIG. **1**.

[0034] At **510**, a Wi-Fi network including a plurality of smart circuit breakers is formed.

[0035] At **520**, a master smart circuit breaker is commissioned to the energy control website in an IoT Hub in a cloud via a mobile energy control application disposed in a mobile user device.

[0036] At **530**, slave smart circuit breakers are simultaneously commissioned to the energy control website in the IoT Hub in the cloud via the mobile energy control application and the master smart circuit breaker.

[0037] At **540**, the mobile energy control application connects all of the smart circuit breakers to the AP, which in turn connects the smart circuit breakers to the energy control website in the cloud.

[0038] FIG. **6** is a flow-chart for a method **600** of automatically commissioning a replacement smart circuit breaker in a Wi-Fi network including a plurality of smart circuit breakers installed in a power distribution system in accordance with a non-limiting embodiment of the disclosed concept. Each smart circuit breaker is structured to be coupled to a power source and respective loads and provide circuit protection in a fault event, the plurality of breaker having been commissioned simultaneously to an energy control website in an IoT Hub in a cloud upon forming the Wi-Fi network. The method **600** may be performed by the components of the system **100** of FIG. **1**.

[0039] At **610**, it is determined whether the Wi-Fi network includes a faulty smart circuit breaker. If yes, the method **600** proceeds to **620**. If no, the method **600** proceeds to **690**.

[0040] At **620**, in response to a determination that the Wi-Fi network includes a faulty smart circuit

breaker, a replacement smart circuit breaker for the faulty smart circuit breaker is installed.

[0041] At **630**, a user selects a master smart circuit breaker from the commissioned smart circuit breakers and actuating the master smart circuit breaker. The user selects an active commissioned smart circuit breaker to act as a master smart circuit breaker and actuates a switch of the master smart circuit breaker to initiate Wi-Fi commissioning procedure of the replacement smart circuit breaker. As such, the mobile energy control application is able to select any smart circuit breaker with active cloud connection and add the replacement smart circuit breaker into the existing setup. The master smart circuit breaker begins ESP-NOW responder and the replacement smart circuit breaker begins ESP-NOW initiator.

[0042] At **640**, the master smart circuit breaker initiates the automatic commissioning of the replacement smart circuit breaker. The master smart circuit breaker scans for the replacement circuit breaker using ESP-NOW protocol.

[0043] At **650**, the replacement smart circuit breaker authenticates the master smart circuit breaker and transmits its device ID (i.e., MAC address including a block ID and a device ID) to the master smart circuit breaker.

[0044] At **660**, the master smart circuit breaker transmits Wi-Fi credentials of an access point (AP) for the Wi-Fi network. The master smart circuit breaker may also send an updated device ID list of all of the smart circuit breakers in the Wi-Fi network to the mobile energy control application. The mobile energy control application then registers the replacement smart circuit breaker and creates an updated list of the smart circuit breakers within its detection range (e.g., without limitation, 2-5 meters).

[0045] At **670**, the replacement smart circuit breaker receives and authenticates the Wi-Fi credentials of the AP.

[0046] At **680**, upon authentication, the replacement smart circuit breaker connects to the AP and then to the IoT Hub in the cloud via the AP. The replacement smart circuit breaker appears on the mobile energy control application and the user configures the replacement smart circuit breaker.

[0047] At **690**, the power distribution system performs normal operation. The power distribution system includes the replacement smart circuit breaker, if any.

[0048] While specific embodiments of the invention have been described in detail, it will be appreciated by those skilled in the art that various modifications and alternatives to those details could be developed in light of the overall teachings of the disclosure. Accordingly, the particular arrangements disclosed are meant to be illustrative only and not limiting as to the scope of disclosed concept which is to be given the full breadth of the claims appended and any and all equivalents thereof.

Claims

1. A method of simultaneously commissioning a plurality of smart circuit breakers to an energy control website in an IoT (Internet-of-Things) Hub in a cloud, each smart circuit breaker being structured to be coupled to a power source and respective load and provide circuit protection in a fault event, comprising: forming, by the plurality of smart circuit breakers, a Wi-Fi network, each smart circuit breaker being capable of acting as a master smart circuit breaker; commissioning a master smart circuit breaker to the energy control website in the IoT Hub via a mobile energy control application disposed in a mobile user device; simultaneously commissioning slave smart circuit breakers to the energy control website in the cloud via the mobile energy control application and the master smart circuit breaker; and connecting the master smart circuit breaker and the slave smart circuit breakers to the IoT Hub via an access point (AP).

2. The method of claim 1, wherein the forming, by the plurality of smart circuit breakers, a Wi-Fi network comprises: installing the plurality of smart circuit breakers in a power distribution system; actuating a switch on one smart circuit breaker to commence a Wi-Fi network forming procedure;

transmitting, by the one smart circuit breaker, a request to connect; actuating switches on remaining smart circuit breakers; transmitting, by the remaining smart circuit breakers, responses to the one smart circuit breaker; and forming, by the one smart circuit breaker and the remaining smart circuit breakers, the Wi-Fi network based on the responses.

3. The method of claim 2, wherein the remaining smart circuit breakers transmit respective device IDs to the one smart circuit breaker.

4. The method of claim 3, wherein the device IDs are MAC (media-access-control) addresses of the remaining smart circuit breakers.

5. The method of claim 2, wherein the Wi-Fi network forming procedure is performed using ESP-NOW protocol.

6. The method of claim 1, wherein the commissioning a master smart circuit breaker comprises: selecting a smart circuit breaker to act as the master smart circuit breaker; actuating a switch on the master smart circuit breaker and placing the master smart circuit breaker in a Bluetooth Low Energy (BLE) mode; scanning, by the mobile energy control application, for presence of smart circuit breakers in detection range thereof in the BLE mode; connecting, by the mobile energy control application, to the master smart circuit breaker via a BLE connection; authenticating Wi-Fi credentials of the AP; connecting, by the mobile energy control application, the energy control website in the IoT Hub to the AP based on the authentication; determining, by the mobile energy control application, that the energy control website in the IoT Hub is successfully connected to the AP; and commissioning the master smart circuit breaker based on the determination that of the energy control website in the IoT Hub is successfully connected to the AP.

7. The method of claim 6, wherein the authenticating the Wi-Fi credentials of the AP comprises: scanning, by the master smart circuit breaker, Wi-Fi APs in vicinity thereof; transmitting, by the master smart circuit breaker, a list of the Wi-Fi APs to the mobile energy control application; and authenticating, by the mobile energy control application, the AP from the list of Wi-Fi APs.

8. The method of claim 7, wherein the Wi-Fi credentials include an SSID and a password for the AP.

9. The method of claim 8, wherein the authenticating, by the mobile energy control application, the AP from the list of Wi-Fi APs includes: selecting the SSID of the AP; and entering the password for the AP in the mobile energy control application.

10. The method of claim 6, wherein the commissioning a master smart circuit breaker further comprises: registering, by the mobile energy control application, the plurality of smart circuit breakers to the energy control website in the IoT Hubs; and configuring the master smart circuit breaker via the mobile energy control application.

11. The method of claim 10, wherein the registering, by the mobile energy control application, the plurality of smart circuit breakers to the energy control website in the IoT Hub includes: creating and transmitting, by the master smart circuit breaker, a list of device IDs of the plurality of smart circuit breakers; and receiving, by the mobile energy control application, the list of device IDs of the plurality of smart circuit breakers.

12. The method of claim 10, wherein the BLE connection is turned OFF upon configuring the master smart circuit breaker via the mobile energy control application.

13. The method of claim 6, wherein in response to a determination that the energy control website is not successfully connected to the AP, the master smart circuit breaker resets and sends a reset command to the slave smart circuit breakers via ESP-NOW protocol.

14. The method of claim 1, wherein the simultaneously commissioning slave smart circuit breakers comprises: receiving, by the slave smart circuit breakers, the Wi-Fi credentials of the AP from the master smart circuit breaker; and establishing, by the slave smart circuit breakers, a connection to the IoT Hub using a connection string and the Wi-Fi credentials via the mobile energy control application.

15. The method of claim 14, wherein the simultaneously commissioning slave smart circuit

breakers further comprises: configuring the slave smart circuit breakers via the mobile energy control application.

16. The method of claim 14, wherein the slave smart circuit breakers are not commissioned to the energy control website in the IoT Hub at one slave smart circuit breaker at a time.

17. A method of automatically commissioning a replacement smart circuit breaker in a Wi-Fi network including a plurality of smart circuit breakers structured to be coupled to a power source and respective loads and provide circuit protection in a fault event, the plurality of breaker having been commissioned simultaneously to an energy control website in an IoT Hub in a cloud upon forming the Wi-Fi network, the method comprising: determining that the Wi-Fi network includes a faulty smart circuit breaker; in response to the determination that the Wi-Fi network includes a faulty smart circuit breaker, installing a replacement smart circuit breaker for the faulty smart circuit breaker; selecting a master smart circuit breaker from the commissioned smart circuit breakers and actuating the master smart circuit breaker; initiating, by the master smart circuit breaker, the automatic commissioning of the replacement smart circuit breaker; authenticating, by the replacement smart circuit breaker, the master smart circuit breaker and transmitting a device ID thereof to the master smart circuit breaker; transmitting, by the master smart circuit breaker, Wi-Fi credentials of an access point for the Wi-Fi network; authenticating, by the replacement smart circuit breaker, the Wi-Fi credentials; and connecting, by the replacement smart circuit breaker, to the AP and to the energy control website in the IoT using the Wi-Fi credentials.

18. The method of claim 17, wherein initiating, by the master smart circuit breaker, the automatic commissioning of the replacement smart circuit breaker includes scanning for the replacement circuit breaker.

19. The method of claim 18, wherein the scanning for the replacement circuit breaker is performed using ESP-NOW protocol.

20. A system, comprising: a plurality of smart circuit breakers structured to be coupled to a power source and respective loads and provide circuit protection during a fault event; a mobile energy control application disposed in a mobile user device; an IoT (Internet-of-Things) Hub disposed in the cloud and communicatively coupled to the mobile energy control application, the IoT Hub including the energy control website; and an access point (AP) structured to be connected to the plurality of smart circuit breakers and the IoT Hub, wherein the plurality of smart circuit breakers are commissioned to the energy control website using a simultaneous Wi-Fi commissioning procedure comprising: forming, by the plurality of smart circuit breakers, a Wi-Fi network, each smart circuit breaker being capable of acting as a master smart circuit breaker; commissioning a master smart circuit breaker to the energy control website in the IoT Hub via the mobile energy control application; simultaneously commissioning slave smart circuit breakers to the energy control website via the mobile energy control application and the master smart circuit breaker; and connecting the master smart circuit breaker and the slave smart circuit breakers to the AP and to the energy control website in the IoT Hub via the AP.
